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releasing oxygen. In fact, in our lungs all the three species coexist.

Few chemical reactions proceed in only one direction whereas many reactions proceed in both the directions and these reactions are called reversible reactions.

In chemical reactions, the concentration of the reactants decreases and that of the products increases with time. In reversible reactions, initially the reaction proceeds towards the formation of the product. Upon formation of the product, the reverse reaction begins to take place. At a particular stage, the rate of the reverse reaction is equal to that of the forward reaction indicating a state of equilibrium.

It is desirable to know the three crucial aspects of chemical reactions namely the feasibility, the rate of the reaction and the extent of reaction. We know that the feasibility of a reaction is given by thermodynamics. Chemical kinetics will tell about the rate of the reaction. The equilibrium constant tells about the extent of a reaction which we will discuss in this chapter. We will also discuss the types of equilibrium, the significance of equilibrium constant and its relationship to thermodynamic quantities and the response of chemical equilibrium to change in the reaction conditions.

8.2. Physical and chemical equilibrium:

There are different types of equilibrium. For example, if two persons with same weight sit on opposite sides of a see-saw at equal distance from the fulcrum, then the see-saw will be stationary and straight and it is said to be in equilibrium.

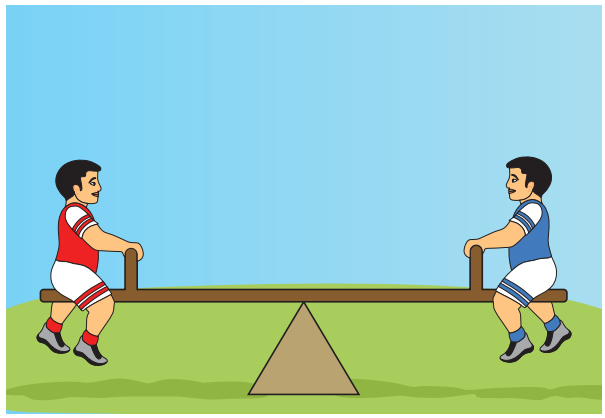


Fig. 8.1 Illustration of equilibrium

Another example of a state of equilibrium is the game of "tug-of-war." In this game a rope is pulled taut between two teams. There may be a situation when both the teams are pulling the rope with equal force and the rope is not moving in either direction. This state is said to be in equilibrium.



Fig. 8.2. Tug-of War

In reversible processes, the rate of two opposing reactions equals at a particular stage. At this stage the concentration of reactants and products do not change with time. This condition is not static and is dynamic, because both the forward and reverse reactions are still occurring with the same rate.